

Hyeonjae Kim

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Research Officer at the Agency for Defense Development

Education

Kyungpook National University (KNU)	Daegu, Republic of Korea
Bachelor of Science in Physics	Mar. 2020 – Feb. 2024
Overall GPA: 4.11/4.3 (3.95/4.0) Major GPA (71 credits): 4.19/4.3 (3.99/4.0)	

Research Experiences

Agency for Defense Development (ADD)	Daejeon, Republic of Korea
Research Officer (Supervisors: Dr. Sin Hyuk Yim, Dr. Sangkyung Lee)	Jun. 2024 – May 2027 (expected)
Research in atomic physics and quantum sensing	
• Designed and implemented an automated absorption spectroscopy system to characterize rubidium vapor cell properties • Built and optimized a zero-field (SERF) optically pumped magnetometer for magnetic noise measurements • Conducted magnetic-anomaly detection experiments and data analysis using a portable single-beam optically pumped magnetometer	

Novel Applied Nano Optics Lab	Daegu, Republic of Korea
Research Intern (Advisor: Prof. Junyeob Yeo)	Jun. 2021 – Jun. 2022
Project: Fabrication of a flexible photo-electrochemical cell using laser	

High Energy Physics Lab (Moon Lab)	Daegu, Republic of Korea
Research Intern (Advisor: Prof. Chang-Sung Moon)	Jun. 2020 – Feb. 2021
Developed momentum reconstruction algorithms for silicon detector simulations	

Publications

Magnetic anomaly detection of an undersea tunnel using a drone-mounted all optical pulsed atomic magnetometer
Manuscript in preparation
H. Kim, S. H. Yim*, S. Lee, E. S. Bang, S. R. Yeo, Y. Kim, T. Jeong, and Y. Lim

Novel sealing of glass-blown atomic gas cells using resistive heating wires
Manuscript under review
S. H. Hong, Y. Lim*, S. H. Yim, S. Lee, H. Kim, T. Jeong, and J. B. Nam

Aging test of an atomic vapor cell with Al₂O₃ wall coating on cubic glass
Applied Optics **64**, 7932-7937 (2025)
H. Kim, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim*

Selected Conference Presentations

Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer
APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting, 2026 (scheduled)

Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells
KISTI-SNU Joint workshop (QNRC Joint Workshop), 2026 (invited)

Towards the development of a SERF magnetometer

Korean Physical Society (KPS) Fall Meeting, 2025

Lifetime extension of rubidium vapor cells by Al_2O_3 coating

APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting, 2025

Linear absorption spectroscopy in rubidium vapor cells: Applications in buffer gas measurement and lifetime estimation

Korean Physical Society (KPS) Spring Meeting, 2025

Honors and Awards

Physics Alumni Association Award and Scholarship (Outstanding Graduate)	Feb. 2024
Department of Physics Alumni Association, Kyungpook National University	
2nd Prize, 2023 MiliTECH Challenge	Dec. 2023
Hosted by KAIST; Awarded by the Agency for Defense Development	
2nd Prize, 2022 MiliTECH Challenge	Dec. 2022
Hosted by KAIST; Awarded by the Korea Military Academy (KMA)	
3rd Prize of Academic Conference	Dec. 2021
College of Natural Sciences, Kyungpook National University	
Dean's List: 3 semesters (2021-1, 2021-2, 2022-2)	Nov. 2021, Apr. 2022, Apr. 2023
College of Natural Sciences, Kyungpook National University	

Grants and Scholarships

National Science and Technology Scholarship	Mar. 2022 – Feb. 2024
Korea Student Aid Foundation (KOSAF)	
Academic Merit Scholarship (Hyoseok Scholarship)	Apr. 2022
Kyungpook National University Alumni Association	
Academic Excellence Scholarship	Aug. 2020, Feb. 2021, Aug. 2021
Kyungpook National University	

Other Research Outputs

Buffer Gas Pressure Estimation for OPAMs	Dec. 2024
Registered software; Korea Copyright Commission (C-2024-047156)	
Rubidium Vapor Cell Lifetime Monitoring and Analysis Tool	Dec. 2024
Registered software; Korea Copyright Commission (C-2024-054414)	

Additional Experiences

Research Officer for National Defense (ROND)	Jun. 2024 – May 2027
First Lieutenant in the Republic of Korea Air Force	
Selected as one of 20 research officers nationwide for STEM-based national defense research	
Teaching Assistant, Classical Mechanics II (PHYS212)	Sep. 2023 – Dec. 2023
Teaching Assistant, Electromagnetism II (PHYS312)	Mar. 2023 – Jun. 2023
International Student Tutor	Feb. 2022 – May. 2022

Skills

Programming: Python, Wolfram Language (Mathematica)

Software: LabVIEW, SolidWorks, COMSOL Multiphysics